Mark schemes

Q1.

(a) (i) $T = 6 \times 9.8 = 58.8$ N

Use of tension being equal to the weight
Accept $6g$

B1

1

(ii) $58.8 = T + 4 \times 9.8$

Three term equation for equilibrium containing 58.8, T and 4×9.8 or equivalent terms.
For M1, 58.8 can be replaced by candidates answer to part (a)(i) provided it is not zero.

M1

Correct equation

A1

$T = 58.8 - 39.2$
$= 19.6$ N

Correct tension
Accept $2g$

A1

3

(b) $6g - T = 6a$

Three term equation of motion for 6 kg particle containing 58.8 or $6g$, T and $6a$.

M1

Correct equation

A1

$T - 4g = 4a$

Three term equation of motion for 4 kg particle containing 39.2 or $4g$, T and $4a$.

M1

Correct equation

A1

$2g = 10a$

$a = 1.96$ m s^{-2}

Correct acceleration
Candidates who work consistently to obtain $a = -1.96$ gain full marks.

A1

5

Special Case for whole system
$6g - 4g = 10a$

Difference in weights equal to 10a

(M1)

A1: Correct equation

(A1)

$a = 1.96$

A1: Correct acceleration

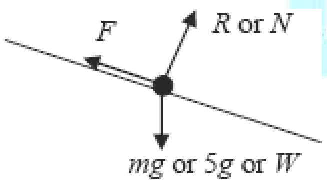
(A1)

(3)

[9]

Q2.

(a)



Correct force diagram with labels and arrows
Accept components of the weight if shown in a different notation with the weight also shown.
B0 if components are shown instead of the weight.

B1

1

(b) (R =) 5 × 9.8cos40° = 37.5 N **AG**

Attempt at resolving perpendicular to the slope (eg 49sin40°)

M1

Correct value from correct working

A1

2

(c) $5 \times 0.8 = 5 \times 9.8\sin40° - \mu \times 5 \times 9.8\cos40°$

Use of $F = \mu R$ at any stage and with any F but with R = 37.5 OE

B1

Three term equation of motion from resolving parallel to the slope with weight component, friction and ma term.

M1

Correct terms seen (may be as 31.5, 37.5μ (or F) and 4)

A1

Correct signs

A1

$$\mu = \frac{5 \times 9.8\sin 40° - 5 \times 0.8}{5 \times 9.8\cos 40°} = 0.733$$

Solving for μ

m1

A1: Correct value for μ

Allow 0.732 but not $\frac{11}{15}$ unless converted to a decimal

A1

6

(d) There is less friction so the coefficient of friction must be less.

Less friction

B1

Smaller coefficient of friction
If the answer and explanation contradict each other, award no marks

B1

2

[11]

Q3.

(a) $a = \frac{dv}{dt} = 12t + 4$

M1 A1

2

(b) Using **F** = m**a**,
Force = 3 × (12t + 4)

M1

When t = 4, force = 3 (12 × 4 + 4)
Force = 156 N

A1

2

[4]

Q4.

(a) Using $F = ma$

$-0.05mv = m\frac{dv}{dt}$

$\therefore \frac{dv}{dt} = -0.05v$

Need to see m terms

B1

1

(b) $\int \frac{dv}{v} = -\int 0.05 \, dt$

B1

$\ln v = -0.05t + c$
$v = Ce^{-0.05t}$

Need first 2 terms

M1

When t = 0, v = 20
$\therefore C = 20$
$v = 20e^{-0.05t}$

only for fully correct solutions

M1A1

4

(c) When v = 10, $10 = 20e^{-0.05t}$

M1

$e^{0.05t} = 2$

A1

$\therefore t = \frac{1}{0.05} \ln 2$

= 13.9 or 20ln2

A1

3

(d) Work done by force is change in KE of car

M1

$= \frac{1}{2}m(20)^2 - \frac{1}{2}m(10)^2$

A1

= 150 m

A1

3

[11]

Q5.

(a) $a = \frac{dv}{dt} = 12t + 4$

M1A1

2

(b) Using $F = ma$,
Force = 3 × (12t + 4)

M1

When t = 4, force = 3 (12 × 4 + 4)
Force = 156 N

A1

2

(c) $r = 2t^3 + 2t^2 - 7t + c$

M1A1

When t = 0, r = 5, $\therefore$ c = 5

M1

$\therefore$ $r = 2t^3 + 2t^2 - 7t + 5$

SC3 if no '+c' seen

A1

4

[8]

Q6.

(a) Using $F = ma$

$-0.05mv = m\dfrac{dv}{dt}$

$\therefore \dfrac{dv}{dt} = -0.05v$

Need to see m terms

B1

1

(b) $\int \dfrac{dv}{v} = -\int 0.05\, dt$

B1

$\ln v = -0.05t + c$

$v = Ce^{-0.05t}$

Need first 2 terms

M1

When $t = 0$, $v = 20$

$\therefore C = 20$

$v = 20e^{-0.05t}$

fully correct solutions

M1A1

4

(c) When $v = 10$, $10 = 20e^{-0.05t}$

M1

$e^{0.05t} = 2$

A1

$\therefore t = \dfrac{1}{0.05}\ln 2$

$= 13.9$

Accept 20 ln 2

A1

3

[8]

Q7.

(a) $900 \times 0.8 = T - 800$

Three term equation of motion for the caravan

M1

Correct equation

A1

$T = 720 + 800 = 1520$ N

Correct result from correct working; AG

A1

3

(b) $2400 \times 0.8 = F - 400 - 800$

Four term equation of motion for the combined body or car

M1

Correct equation

A1

$F = 1920 + 1200 = 3120$ N

Correct force

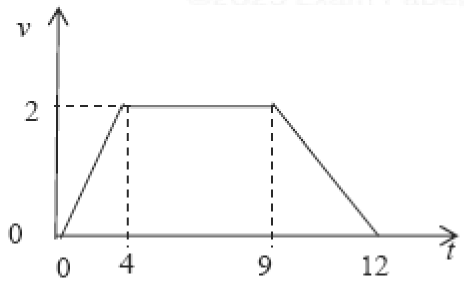
A1

3

[6]

Q8.

(a)



Starts and finishes at rest

B1

Correct shape

B1

Correct values on t-axis

B1

Correct values on v-axis

B1

Condone omission of the origin

4

(b) $s = \frac{1}{2}(5 + 12) \times 2$

or $s = \frac{1}{2} \times 2 \times 4 + 5 \times 2 + \frac{1}{2} \times 2 \times 3 = 17$

Use of the area under the graph (or equivalent) to find s

M1

= 17

Correct distance
***SC** When 21 used instead of 12 allow full marks for s = 26*

A1

2

(c) max $a = \frac{2}{4} = 0.5$

Maximum acceleration

B1

$300 \times 0.5 = T - 300 \times 9.8$

Three term equation of motion using their a

M1

Correct equation using a = 0.5

A1

$T = 2940 + 150 = 3090$

Correct tension

A1

4

[10]

Q9.

(a) The string is light and inextensible or inelastic or taut

First assumption

B1

Second assumption

B1

2

(b) $6 = 0 + 4a$

Finding a using a CA equation

M1

$a = \frac{6}{4} = 1.5$

Correct a from correct working

A1

2

(c) $7 \times 9.8 - T = 7 \times 1.5$

Three term equation of motion with F for the 7 kg particle. Correct equation

M1A1

$T = 68.6 - 10.5 = 58.1$

Correct tension

A1

3

(d) $58.1 - F = 13 \times 1.5$

Three term equation of motion with F for the 13 kg particle. Correct equation

M1A1

$F = 58.1 - 19.5 = 38.6$

Correct F

A1

$R = 13.98 = 127.4$

Correct R

B1

$38.6 = \mu \times 127.4$

Use of $F = \mu R$

dM1

$\mu = \frac{38.6}{127.4} = 0.303$

Correct coefficient of friction

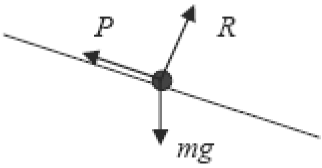
A1

6

[13]

Q10.

(a)



Correct diagram with arrows and labels
Must not use F instead of P
Condone resistance instead of P

B1

1

(b) P = 100 × 9.8 sin 4°

Resolving weight (must see 100)

M1

Using sin 4° or cos 86°

M1

= 68.4

***AG** Correct P from correct working*

A1

3

(c) 100a = 100 × 9.8 sin 5° − 100 × 9.8 sin 4°

Three term equation of motion

M1

Weight resolved correctly

A1

Correct equation

A1

$a = \frac{100 \times 9.8 \sin 5° - 100 \times 9.8 \sin 4°}{100}$

= 0.171

Correct a. (Accept 0.170 or 0.17)

A1

4

(d) You would expect P to vary with the speed of the car.

Correct explanation

B1

1

[9]

Q11.

(a) $320 = \frac{1}{2} \times a \times 80^2$

Use of constant acceleration equation with $u = 0$

M1

Correct equation

A1

$$a = \frac{2 \times 320}{80^2} = 0.1\,\text{m s}^{-2}$$

AG Correct acceleration from correct working

A1

3

(b) $v = 0 + 0.1 \times 80$

Use of constant acceleration equation with $u = 0$

M1

$= 8\ \text{m s}^{-1}$

Correct velocity

A1

2

(c) $L - 450 \times 9.8 = 450 \times 0.1$

Three term equation of motion

M1

Correct equation

A1

L = 45 + 4410
= 4455
= 4500 N (to 3sf)

AG Correct force

A1

3

(d) 4410 N

Correct force

B1

1

[9]

Q12.

(a) $3.45g - T = 3.45a$

Three term equation of motion for one particle

M1

Correct equation

A1

$T - 1.45g = 1.45a$

Three term equation of motion for other particle

M1

Correct equation

A1

$2g = 4.9a$

$$a = \frac{2 \times 9.8}{4.9} = 4 \text{ m s}^{-2}$$

AG Correct acceleration from correct working

A1

5

(b) $T = 1.45 \times 4 + 1.45 \times 9.8$

Use of one equation from (a) to find T

M1

= 20.01
= 20.0 N (to 3 sf)

Correct T

A1

2

(c) $s = \frac{1}{2} = 0.5$ m

Use of $s = \frac{1}{2}$

M1

$v^2 = 2 \times 4 \times 0.5$

Use of constant acceleration equation with $u = 0$

M1

$v = 2$ m s^{-1}

Correct speed
(no negative sign)

A1

3

[10]

Q13.

(a) $T - 800 = 1200 \times 0.4$

Three term equation of motion for the car

M1

Correct equation

A1

$T = 800 + 480$
$= 1280$ N

Correct tension

A1

Treat calculation of two tensions as two methods unless one selected
Treat sum or difference of two tensions as an incorrect method

3

(b) $3000 - 800 - F = 4000 \times 0.4$

Four term equation of motion (truck or both)

M1

Correct terms

A1

Correct signs

A1

$F = 3000 - 800 - 1600$
$F = 600$ N

AG Correct resistance force from correct working

A1

OR

$3000 - 1280 - F = 2800 \times 0.4$
$F = 3000 - 1280 - 1120$
$F = 600$ N

4

(c) Increase, because a greater tension would be needed so that the horizontal component would be the same as the tension above.

Greater

B1

Reason
Second B1 dependent on the first B1 mark

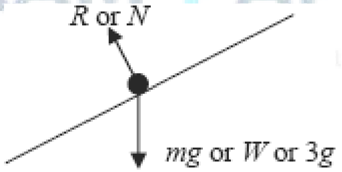
B1

2

[9]

Q14.

(a) (i)



Correct diagram with arrows and labels

B1

1

(ii) $3a = 3g \sin 30°$

Two term equation of motion

M1

$a = g \sin 30° = 4.9 \text{ m s}^{-2}$

AG Correct acceleration from correct working (Allow $a = g$ sin 30°)

A1

2

(b) (i) $5 = \frac{1}{2} a \times 2^2$

Constant acceleration equation with u = 0

M1

a = 2.5 m s^{-2}

AG Correct answer from correct working.
(Use of v = 5 must be justified)

A1

2

(ii) 3 × 2.5 = 3g sin 30° – F

Three term equation of motion

M1

Correct equation

A1

F = 3g sin 30° – 7.5
= 7.20 N (to 3 sf)

Correct F
Accept 7.2 N

A1

3

(iii) R = 3g cos 30° (= 25.46)

Resolving perpendicular to the slope to find R

M1

Correct R

A1

7.2 = μ × 3g cos 30°

Use of $F = \mu R$

M1

Correct expression

A1F

$$\mu = \frac{7.2}{3g\cos 30°} = 0.283$$

Correct μ
Accept 0.282
(Follow through from incorrect F from above, but not an incorrect R)

A1F

5

(iv) Reduce a, as the air resistance would reduce the magnitude of the resultant force or because the air resistance increases as the velocity increases towards its terminal value

Reduces

B1

Explanation
Second B1 dependent on the first B1 mark

B1

2

[15]

Q15.

(a) Using $F = ma$:
$2400\mathbf{i} - 4800t\mathbf{j} = 800\mathbf{a}$

M1

$\mathbf{a} = 3\mathbf{i} - 6t\mathbf{j}$

A1

2

(b) $\mathbf{v} = \int \mathbf{a} \, \mathrm{d}t$

M1

$= 3t\mathbf{i} - 3t^2\mathbf{j} + \mathbf{c}$

*Condone no '+ **c**'*

A1

When $t = 0$, $\mathbf{v} = 6\mathbf{i} + 30\mathbf{j}$
$\therefore$ $\mathbf{c} = 6\mathbf{i} + 30\mathbf{j}$

*Needs '+ **c**' above*

M1

$\therefore$ $\mathbf{v} = (3t + 6)\mathbf{i} + (30 - 3t^2)\mathbf{j}$

AG

A1

4

(c) $\mathbf{r} = \int \mathbf{v}\ \mathrm{d}t$

M1

$= (\frac{3}{2}t^2 + 6t)\,\mathbf{i} + (30t - t^3)\,\mathbf{j} + \mathbf{d}$

*A1 **i** term, A1 **j** term; condone no '+ **d**'*

A1,A1

When $t = 0$, $\mathbf{r} = 2\mathbf{i} + 5\mathbf{j}$

$\therefore\ \mathbf{d} = 2\mathbf{i} + 5\mathbf{j}$

M1

$\therefore\ \mathbf{r} = (\frac{3}{2}t^2 + 6t + 2)\mathbf{i} + (30t - t^3 + 5)\mathbf{j}$

A1

5

[11]

Q16.

(a) Using $F = ma$:

$-\lambda mv = ma = m\frac{\mathrm{d}v}{\mathrm{d}t}$

Condone no '–'

M1

$\therefore \frac{\mathrm{d}v}{\mathrm{d}t} = -\lambda v$

AG

Note: no use of $m \Rightarrow$ no marks in (a)

A1

2

(b) $\int \frac{\mathrm{d}v}{v} = -\lambda \int \mathrm{d}t$

M1

In $v = -\lambda t + c$

$v = C\,e^{-\lambda t}$

Needs '+ c'

A1

When t = 0, $v = U \Rightarrow C = U$

Needs correct working

M1

$v = U\,e^{-\lambda t}$

AG

A1

4

[6]

Q17.

(a) (i) T = 0.6 × 9.8 = 5.88N or 0.6g

B1

1

(ii) Force = 2T = ↓ 11.76N or 11.8N or 1.2g

B1

2

Magnitude

B1

direction

B1

(b) (i) Q:0.8g – T = 0.8a

M1

T – 0.6g = 0.6a

either equation

A1

0.2g = 1.4a

A1

***alt** M1A1 if solving for T, full method for solving, A1 accurate attempt*

m1

a = 1.4a

A1

T = 6.72N

CAO

***S.C** whole string.*

To find a: 0.2g = 1.4a (M1)

a = 1.4 A1

To find T: (M1A1)

CAO

A1

6

(ii) Force = 2T = 13.44N

B1

1

[10]

Q18.

(a) (i) R = 80cos 25°

component attempted

M1

correct component

A1

R = 72.5N

CAO

A1

3

(ii) $F = 0.32 \times 72.5$

condone inequality

M1

F = 23.2N

CAO

A1

2

(iii) $T + F$ = 80cos65°

3 forces direction correct, component

M2

attempted

A1

T = 10.6N

ft friction

A1ft

4

(iv) $T = F + 80\cos 65°$

3 forces, direction correct, component

M1

attempted

A1

T = 57.0N (57N)

ft friction

A1ft

3

(iv) Mass = $\frac{80}{g}$ = (8.16kg)

B1

1

(b) 80 cos 65° – F = mass × acceleration

3 terms, component attempted

M1

10.6 = $\frac{80}{g}$ × acc

all correct

A1

acc = 1.30 ms^{-2} (1.3 ms^{-2})

CAO

A1

3

[16]

Q19.

1600 $\frac{dv}{dt}$ = –40v

Applying Newton's second law with 40v and $\frac{dv}{dt}$

M1

Correct equation

A1

$\int \frac{1}{v} \mathrm{d}v = \int -\frac{1}{40} \mathrm{d}t$

Separating variables

dM1

$\ln v = -\frac{t}{40} + c$

integrating to get lnv term.

dM1

$v = Ae^{-\frac{t}{40}}$

Correct integral with or without c

A1

$t = 0$, $v = 20 \Rightarrow c = 20$

Finding constant

dM1

$v = 20e^{-\frac{t}{40}}$

Correct final result

A1

[7]

Q20.

(a) $\mathbf{F} = 12\cos t\mathbf{i} - 30\sin t\mathbf{j}$

Use of **F** = *m***a**

M1

$\mathbf{F}(0) = 6 \times 2\mathbf{i}$ so $F = 12$ N

Correct **F**

A1

Correct magnitude

A1

3

(b) $\mathbf{v} = \int 2\cos t\,dt\,\mathbf{i} + \int -5\sin t\,dt\,\mathbf{j}$

Integrating

M1

$= (2\sin t + c_1)\mathbf{i} + (5\cos t + c_2)\mathbf{j}$

Correct **i** *component*

A1

Correct **j** *component*

A1

Both of above with or without constants

$\mathbf{v}(0) = 2\mathbf{i} + 10\mathbf{j} \Rightarrow c_1 = 2,\ c_2 = 5$

Finding constants of integration

dM1

$\mathbf{v} = (2\sin t + 2)\mathbf{i} + (5\cos t + 5)\mathbf{j}$

Correct final answer

A1

5

[8]

Q21.

(a) The boat is not affected by the movement of the water.

First assumption

B1

The resistance force will directly oppose the motion of the boat and be the only force that needs to be considered.

second assumption

B1

2

(b) The boat is a particle. / There is no wind. / No air resistance.

Appropriate assumption

B1

1

(c) (i) $80\frac{dv}{dt} = -20v$

Use of Newton's second law, $\frac{dv}{dt}$ *and 20v.*

M1

$\frac{dv}{dt} = -\frac{v}{4}$

Correct result

A1

2

(ii) $\int \frac{1}{v}\,dv = -\int \frac{1}{4}\,dt$

Sep. of variables and forming integrals

M1

$\ln v = -\frac{t}{4} + c$

Integrating to get an 1n v term

dM1

Correct integrals with or without c

A1

$v = Ae^{-\frac{t}{4}}$

$v = 12$, $t = 0 \Rightarrow A = 12$

Finding A or c

dM1

$v = Ae^{-\frac{t}{4}}$

Correct final result

A1

5

[10]

Q22.

(a) $0.3g - T = 0.3a$

Consistent reversal of signs in both equations 4 marks;

M1A1

$T - 0.2g = 0.2a$

Reversal of signs in one equation, M1 A1 M1 A0

M1A1

$a = 1.96 \text{ ms}^{-2}$

Sign change needs justification
(Whole string:
Equation, 0.3g – 0.2g = 0.5a M1A1,
a = 1.96 A1) max 3/5

A1

5

(b) $0.1g - R = 0.1 \times 1.96$

all terms

M1

all correct

A1

$R = 0.784 \text{ N}$

ft provided R positive

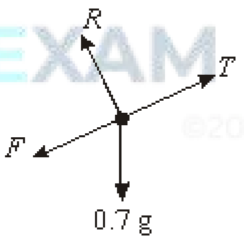
A1F

3

[8]

Q23.

(a)



Accept W or mg (or 6.86) for weight
Arrows and labels needed
(can replace W with 2 correct components)

B1

1

(b) $R = 0.7g \cos 22°$

component of weight attempted

M1

$R = 6.36 \text{ N}$

all correct, including signs

A1

CAO

A1

3

(c) $F = 0.25 \times 6.36$

$F = 1.59\text{N}$

CAO

M1 A1

2

(d) 5.6 – 0.7g sin 22°

a = 2.06 ms^{-2}

4 terms with weight component attempted

M1

A marks -1 each error, accept $\pm 0.7a$

A2

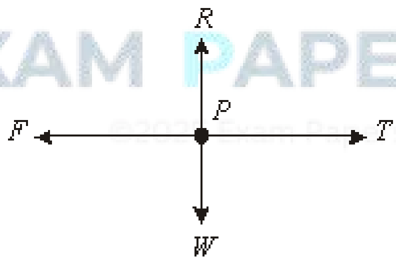
FT one error, accept $\pm$

A1F

[10]

Q24.

(a) (i)



Accept mg, 0.4g or 3.92 for weight
Arrows and labels needed

B1

1

(ii) F = 0.5 = × (0.4 × 9.8)

Need to see 0.4×9.8 *or 3.92 used*

M1

F = 1.96N

A1

2

(b) $T - 1.96 = 1.4a$
$0.3g - T = 0.3a$

M1A1

Consistent reversal of signs in both equations 4 marks; reversal of signs in one equation, M1 A1 M1 A0

M1A1

$a = 1.4 \text{ms}^{-2}$

Sign change needs justification (whole string: equation, $0.3g - 1.96 = 0.7a$ *M1A1* $a = 1.4$ *A1) max 3/5*

A1

5

(c) $v = 14 \times 3$
$v = 4.2 \text{ms}^{-2}$

Full method

M1

CAO

A1

2

(d) P : Friction will cause speed to decrease

M1

Accept decelerate or comes to rest

A1

Q : Gravity will cause speed to increase

M1

Accept accelerate

A1

4

[14]

Q25.

(a) $20\frac{dv}{dt} = -10\sqrt{v}$

applying Newton's second law with $\frac{dv}{dt}$

M1

correct differential equation

A1

$\frac{dv}{dt} = -\frac{\sqrt{v}}{2}$

separating variables

dM1

$\int \frac{1}{\sqrt{v}}\,dv = \int -\frac{1}{2}\,dt$ AG

$2\sqrt{v} = \frac{t}{2} + c$

integrating

dM1

correct integrals

A1

$t = 0$, $v = 25 \Rightarrow c = 10$

finding the constant of integration

$v = \left(\frac{20-t}{4}\right)^2$

correct final result from correct working

A1

7

(b) $t = 20$

correct time

B1

1

[8]

Q26.

(a) $F = 800 + \frac{1200}{20}t = 800 + 60t$

finding the gradient of the line

M1

correct gradient

A1

$1200a = 800 + 60t$

correct intercept

B1

$a = \frac{800}{1200} + \frac{60}{1200}t = \frac{2}{3} + \frac{t}{20}$

using Newton's second law on two terms

dM1

AG

correct result from correct working

A1

5

(b) $v = \int \frac{2}{3} + \frac{t}{20}\,dt = \frac{2t}{3} + \frac{t^2}{40} + c$

integrating

M1

correct integral with or without c

A1

$v = 0, t = 0 \Rightarrow c = 0$

A1

$v = \frac{2t}{3} + \frac{t^2}{40}$

showing c = 0

A1

3

(c) $s = \int_0^{20} \frac{2t}{3} + \frac{t^2}{40}\,dt$

integrating

M1

correct integral, with or without c.

A1

$$= \left[\frac{t}{3} + \frac{t^3}{120}\right]_0^{20}$$

use of both limits or finding c

dM1

= 200 m

correct distance

A1

4

(d) The $\frac{2t}{3}$ term would change, because only

correct term

B1

the constant term in the force would change.
When integrated this becomes the t term in the velocity.

correct explanation

B1

2

[14]

Q27.

(a) (i) $40 = 12 + 100a$

Use of a constant acceleration equation to form equation for a

M1

$a = \frac{40 - 12}{100} = 0.28$ ms^{-2} AG

AG; correct answer from correct working

A1

2

(ii) $s = \frac{1}{2}(12 + 40) \times 100$

Expression for distance, using t =100

M1

= 2600m

Correct final distance

A1

2

(c) $F - 40000 = 200 \times 1000 \times 0.28$

Three term equation of motion

M1

$F = 40000 + 56000 = 96000\ N$

Correct equation

A1

Correct force

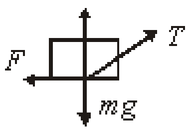
A1

3

[7]

Q28.

(a)



Correct diagram

B1

1

(b) 40 cos 30° – F = 25 × 0.1

F = 40 cos 30° – 2.5 = 32.1 N

Three term equation of motion

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(c) R + 40 sin 30° = 25 × 9.8

R = 225 N

Resolving vertically

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(d) $32.1 = 225\mu$

use of F = μR

M1

$\mu = \frac{32.1}{225} = 0.143$

Correct μ

A1

2

[9]

Q29.

(a) $F = 0.8 \times 8 \times 9.8 = 62.72 \text{ N}$

Use of F = μR *with R = 8 ×9.8*

M1

Correct F

A1

2

(b) $7g - T = 7a$

Three term equation of motion for one particle

M1

$T - 62.72 = 8a$

Correct equation

A1

$7g - 62.72 = 15a$

Three term equation of motion for other particle

M1

$a = \frac{7g - 62.72}{15} = 0.392$

Correct equation

A1

AG; correct a from correct working

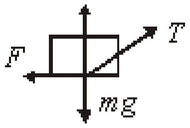
A1

5

[7]

Q30.

(a)



Correct diagram

B1

1

(b) 40 cos 30° – F = 25 × 0.1

F = 40 cos 30° – 2.5 + 32.1 N

Three term equation of motion

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(c) R + 40 sin 30° = 25 × 9.8

R = 225 N

Resolving vertically

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(d) $32.1 = 225\mu$

use of F = μR

M1

$\mu = \dfrac{32.1}{225} = 0.143$

Correct μ

A1

2

(e) Friction will decrease as normal reaction

Decrease in friction

B1

decreases

Normal reaction decreases

B1

2

[11]

Q31.

(a) Light or smooth

Acceptable assumption

B1

1

(b) $5g - T = 5a$

Three term equation of motion for one particle

M1

$T - 2g = 2a$

Correct equation

A1

$3g = 7a$

Three term equation of motion for other particle

M1

$a = \frac{3g}{7} = 4.2\,\text{ms}^{-2}$

Correct equation

A1

AG; correct acceleration from correct working

A1

5

(c) $T = 2 \times 4.2 + 2 \times 9.8 = 28$ N

Substitute a = 4.2 into one equation of motion

M1

Correct tension

A1

2

[8]

Q32.

(a) $s = \frac{1}{2} \times 15 \times 20 = 150$

M1: Finding length of first stage
M1: Finding length of second stage
A1: Both distances correct

M1

$s = \frac{1}{2} \times 15 \times 80 = 600$

M1A1

$s = 600 + 150 = 750$ m

A1: Correct total distance

A1

4

(b) (i) $t = \frac{750}{15} = 50$ s

B1: Correct time or their distance correctly divided by 15

B1ft

1

(ii) Delay = 120 – 50 = 70 s

B1: Correct time of their previous time correctly subtracted from 120 to give a positive answer

B1ft

1

(c) $a = \frac{15}{80} = \frac{3}{16} = 0.1875$ ms^{-2}

M1: Finding acceleration
A1: Correct acceleration

M1A1

$F = 500000 \times 0.1875 = 93800$ N (to 3sf)

M1:Use of $F = ma$
A1: Correct force

M1A1

4

[10]

Q33.

(a) $9g - T = 9a$

M1: Equation for one particle
A1: Correct equation

M1A1

T – 5g = 5a

M1: Equation for other particle
A1: Correct equation

M1A1

$4g = 14a$

$a = \frac{4g}{14} = 2.8 \text{ ms}^{-2}$

A1:Correct a from correct working
AG

A1

5

(b) $T - 5g = 5 \times 2.8$

M1: Substituting acceleration to find T

M1

$T = 63$ N

A1: Correct tension

A1

2

(c) $s = \left(\frac{1}{2} \times 2.8 \times 0.5^2\right)$

M1: Constant acceleration equation with
u = 0 and a ≠ g to find s. Allow ± answers.
A1: Correct equation

M1A1

= 0.35m

A1: Correct distance

A1

Total = 2 × 0.35 = 0.7 m

A1: Doubling their distance to get total distance apart

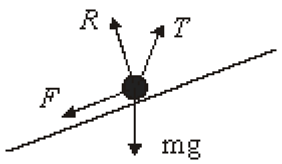
A1ft

4

[11]

Q34.

(a)



B1: Correct force diagram

B1

1

(b) $R + 20 \sin 30° = 6g \cos 10°$

M1: Resolving perpendicular to the slope with 3 terms
A1: Correct equation

M1A1

$R = 6\,g \cos 10° – 20 \sin 30°$

dM1: Solving for R

dM1

$R = 47.9$ N (to 3 sf) AG

A1: Correct R from correct working

A1

4

(c) $F = \mu R$

M1: Use of $F = \mu R$

M1

$6 \times 0.4 = 20\cos 30° – 6g \sin 10° – \mu R$

M1: Resolving parallel to slope to get 4 term equation of motion
A1: Correct equation

M1A1

$\mu R = 4.710$

A1: Correct F / μR

A1

$\mu = \frac{4.710}{47.91} = 0.0983$

dM1: Solving for μ

dM1

A1: AWRT 0.098

A1

6

[11]

Q35.

(a) No air resistance / Only gravity or weight

B1: Acceptable assumption

B1

1

(b) (i) $0.2 \times 8 = 0.2 \times 9.8 - R$

M1: Three term equation of motion
A1: Correct equation

M1A1

$R = 0.36$ N

A1: Correct magnitute of the resistance force

A1

3

(b) (ii) **Increases** as the speed increases

B1: Correct explanation

B1

1

(c) (i) ± 9.8 ms^{-2}

B1:CAO

B1

1

(ii) **Decreases** towards zero

B1: Correct explanation

B1

1

[7]